Chapter 4: Project Integration Management

Information Technology Project Management, Eighth Edition

Note: See the text itself for full citations.

What is Project Integration Management? (1 of 3)

- Project managers must coordinate all of the other knowledge areas throughout a project's life cycle
- Many new project managers have trouble looking at the "big picture" and want to focus on too many details (See opening case for a real example)
- Project integration management is not the same thing as software integration

What is Project Integration Management? (2 of 3)

- Main processes
  - Developing the project charter
  - Developing the project management plan
  - Directing and managing project work
  - Monitoring and controlling project work
  - Performing integrated change control
  - Closing the project or phase

Strategic Planning and Project Selection (1 of 3)

- Strategic planning involves determining long-term objectives
  - Analyzing the strengths and weaknesses of an organization
  - Studying opportunities and threats in the business environment
  - Predicting future trends
  - Projecting the need for new products and services
- SWOT analysis
  - Strengths, Weaknesses, Opportunities, and Threats
- Identifying potential projects
  - Start of project initiation
- Aligning IT with business strategy
  - Organization must develop a strategy for using IT to define how it will support the organization’s objectives

Methods for Selecting Projects

- Potential projects must be narrowed down
  - Methods for selecting projects
    - Focusing on broad organizational needs
    - Categorizing information technology projects
    - Performing net present value or other financial analyses
    - Using a weighted scoring model
    - Implementing a balanced scorecard

Focusing on Broad Organizational Needs

- Projects that address broad organizational needs are much more likely to be successful because they will be important to the organization
  - Examples: improve safety or increase morale
- Important criteria for selecting projects
  - Need
  - Funding
  - Will

Categorizing IT Projects

- Categorizations
  - Respond to a problem, opportunity, or directive
  - How long it will take to do and when it is needed
  - Overall priority of the project

Performing Financial Analyses

- Financial considerations are often an important consideration in selecting projects
  - Regardless of current economics
- Primary methods for determining the projected financial value of projects
  - Net present value (NPV) analysis
  - Return on investment (ROI)
  - Payback analysis

Net Present Value Analysis (1 of 4)

- Method of calculating the expected net monetary gain or loss from a project by discounting all expected future cash inflows and outflows to the present point in time
  - Projects with a positive NPV should be considered if financial value is a key criterion
  - Projects with higher NPVs are preferred

Net Present Value Analysis (2 of 4)

	A	B	C	D	E	F	G
1	Discount rate	10%					
2							
3	PROJECT 1	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
4	Benefits	$0	$2,000	$3,000	$4,000	$5,000	$14,000
5	Costs	$5,000	$1,000	$1,000	$1,000	$1,000	$9,000
6	Cash flow	($5,000)	$1,000	$2,000	$3,000	$4,000	$5,000
7	NPV →	$2,316					
8		Formula =npv(b1,b6:f6)					
9							
10	PROJECT 2	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
11	Benefits	$1,000	$2,000	$4,000	$4,000	$4,000	$15,000
12	Costs	$2,000	$2,000	$2,000	$2,000	$2,000	$10,000
13	Cash flow	($1,000)	$0	$2,000	$2,000	$2,000	$5,000
14	NPV →	$3,201					
15		Formula =npv(b1,b13:f13)					
16							
17							

Note that totals are equal, but NPVs are not because of the time value of money

FIGURE 4-4 Net present value example

Net Present Value Analysis (3 of 4)

Discount rate	**8%**					
Assume the project is completed in Year 0			**Year**			
	0	**1**	**2**	**3**	**Total**	
Costs	140,000	40,000	40,000	40,000		
Discount factor	1	0.93	0.86	0.79		
Discounted costs	**140,000**	**37,200**	**34,400**	**31,600**	**243,200**	
Benefits	0	200,000	200,000	200,000		
Discount factor	1	0.93	0.86	0.79		
Discounted benefits	**0**	**186,000**	**172,000**	**158,000**	**516,000**	
Discounted benefits - costs	(140,000)	148,800	137,600	126,400	**272,800**	**←NPV**
Cumulative benefits - costs	(140,000)	8,800	146,400	272,800		
ROI	**112%**					
	Payback in Year 1					

FIGURE 4-5 JWD Consulting net present value and return on investment example

The mathematical formula for calculating NPV is

$$\mathrm{NPV} = \sum_{t=0\ldots n} A_t/(1+r)^t$$

where t equals the year of the cash flow, n is the last year of the cash flow, A is the amount of cash flow each year, and r is the discount rate.

Monitoring Risks

- Involves ensuring the appropriate risk responses are performed, tracking identified risks, identifying and analyzing new risk, and evaluating effectiveness of risk management throughout the entire project
  - Project risk management does not stop with the initial risk analysis
- Carrying out individual risk management plans involves monitoring risks based on defined milestones and making decisions regarding risks and their response strategies
  - Project teams sometimes use workarounds—unplanned responses to risk events—when they do not have contingency plans in place